Adrian Lara

contact@adrianlara.dev | linkedin.com/in/adrian-lara-cs | github.com/adlune | adrianlara.dev

Education

National Autonomous University of Mexico (UNAM) Morelia, Michoacan

Bachelor of Science in Information Technologies for Science *Aug. 2024 – Present*

- Relevant Coursework: Databases, Descriptive and Inferential Statistics, Linear Algebra, Differential Calculus, Differential Equations, Machine Learning, Neural Networks, Data Structures, Data Mining, Databases, Networks and Telecommunications

Projects

Force-Directed N-Body Network Anomaly Detection | *Python, Docker* Aug. 2026 – Present

- Designing a force-directed N-body simulation to model network traffic as dynamic graphs, enabling novelty detection
- Applying physics-based force modeling to cluster benign traffic patterns and isolate anomalous network behavior across multiple attack categories
- Containerizing the pipeline with Docker for reproducible experimentation

Cybersecurity Home Lab | *Kali Linux, VMWare* 2025 – Present

- Set up a Kali Linux virtual environment for cybersecurity tool experimentation and self-study

Video Game Market Regional Analysis & Prediction | *R, LaTeX* 2026

- Conducted descriptive and inferential statistical analysis of the vgsales dataset, examining video game sales patterns across North America, Europe, Japan, and global
- Performed hypothesis testing and built a linear regression model relating Japan-region sales to global sales, identifying niche demand for RPG titles in the Japanese market
- Extended the analysis in a machine learning project, training and comparing multiple predictive models on the same dataset and complemented the results with the original analysis visualizations
- Made a LaTeX report with data visualizations for both course submissions

PumaRide | *Python, Django, FastAPI, OpenRouteService API, Leaflet* 2026

- Contributed to building a full-stack web application that calculates optimal travel routes across the UNAM ENES Morelia campus and surroundings
- Developed a REST API backend with FastAPI to handle route computation requests via the OpenRouteService API
- Implemented the user-facing frontend with Django, allowing users to input start and destination points and view calculated routes

Technical Skills

Languages: Python, SQL, HTML/CSS, R, Bash
Frameworks: FastAPI, Flask, Django
Libraries: pandas, NumPy, Matplotlib, seaborn, scikit-learn, PyTorch, BeautifulSoup
Developer Tools: Git, Docker, VS Code, Linux (Kali/Ubuntu/Arch)
Security Tools: Nmap, Wireshark, Burp Suite, Metasploit